Joshua Emmanuel A. Guerrero

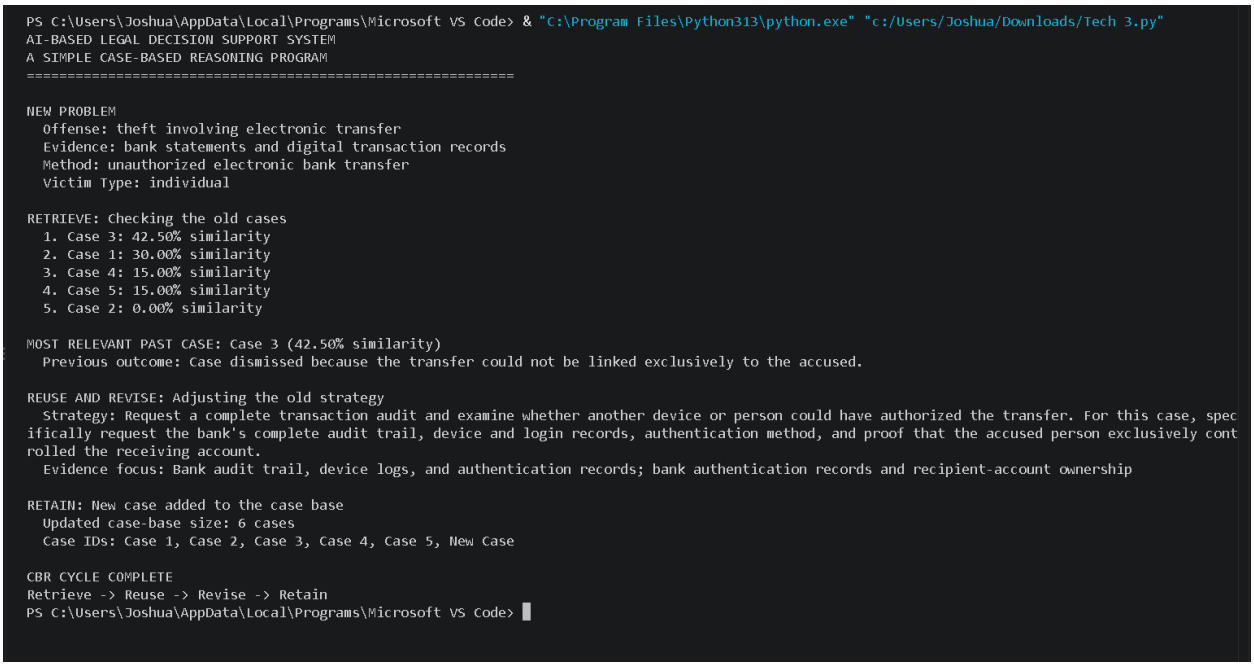
John Patrick Villaraza

Short Documentation

AI-Based Legal Decision Support System Using Case-Based Reasoning

This project develops a simple Case-Based Reasoning system that helps analyze a legal case involving an unauthorized electronic bank transfer. It uses five previous cases and compares them based on the offense, evidence, method used, and victim type, with different weights assigned to each attribute. Case 3 was identified as the closest match with a 42.50% similarity score because it also involved an unauthorized bank transfer. The system reused Case 3's strategy and adapted it by adding checks for bank records, device logs, authentication details, and ownership of the receiving account. Afterward, the new case and its solution were added to the case base, completing the retrieve, reuse, revise, and retain cycle.

So here's the Output of our program



(The code is in TA3 Assessment)

Pair Reflection (Guerrero and Villaraza)

We worked together by creating the legal case base, coding the similarity-scoring process, and testing the program output. One challenge we encountered was deciding which legal details were important when comparing the cases. We also had to make sure that the retrieved solution was adjusted to fit the new electronic-transfer situation. Through this activity, we learned how CBR uses previous cases to help solve new problems. We also learned that retaining new cases can improve the system's knowledge for future comparisons. Overall, the project helped us understand how CBR can support legal decision-making using previous experiences.